Radar Signal Parameter Estimation with Sparse

2014 IEEE International Conference on Computer and Information Technology

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***Abstract*—In this paper, a sparse Bayesian representation model based on zoom-dictionary is proposed to estimate the parameters of the complex radar signal. According to the traditional parameter estimation methods, the parameters in a signal can be estimated via a dictionary whose atoms defined on the discrete grid in the parameter space. Then, we use a Beta-Bernoulli prior to realize the sparse selection of the parameterized atoms in a given dictionary (i.e. achieve the estimation of the parameters), where the dictionary grids can be iteratively zoomed in to estimate the parameters more precisely. The varaitional Bayesian (VB) inference algorithm is developed for the proposed sparse Bayesian model. Furthermore, based on the preliminary parameter estimation, we develop a weighted clustering method to further enhance the accuracy of the parameter estimation. Simulation examples demonstrate that the proposed algorithm can achieve much more accurate parameter estimation results.**

Keywords—Parameter estimation, Sparse Bayesian, Beta-Bernoulli Prior, Variational Bayesian (VB) inference, Clustering

I. INTRODUCTION

Nowadays, radar signal parameter estimation becomes increasingly important in practical applications. When a radar transmits an electromagnetic signal to a target, the signal interacts with the target and returns back to the radar. Changes in the properties of the returned signal reflect the characteristics of interest for the target, which explicitly represent the different values of the parameters in the signal echoes. Therefore, we can obtain more information to analyze the characteristics of the radar target by estimating the parameters of the returned signal.

Recently, Compressive sensing (CS) has attracted much attention in the super resolution and parameter estimation of the radar signal [1]. CS theory [2, 3] reveals that if a signal may be sparsely rendered in some basis, it may be exactly recovered based on a relatively small set of random-projection measurements. To realize the parameter estimation, CS methods model the dictionary as a parameterized dictionary, with a large initial parameters to cover as much as possible the estimated parameters [1, 4]. However, they can just adopt the fixed basis functions with finite parameters set. In other words, the parameter interval is an important point. If the parameter search intervals are too large, it is hard to search for the optimal parameterized atoms, and the sum of some basis functions may be considered as a true signal component; inversely, if search intervals are too small, the computational complexity would increase significantly.

Therefore, a technique named as sparse Bayesian representation with zoom-dictionary (SBRZD) is proposed herein to estimate the radar signal parameters, which joints sparse Bayesian parameter estimation and dictionary refinement [5, 6, 7, 18] based on the variational Bayesian (VB) [9, 10, 11] algorithm. After accomplishing the preliminary parameter estimation, we utilize a weighted clustering to further enhance the accuracy of parameter estimation.

The remainder of the paper is organized as follows: in Section Ⅱ, the signal model and the sparse Bayesian parameter estimation technique with fixed dictionary are briefly introduced. Section Ⅲ develops our proposed parameter estimation algorithm with the zoom-dictionary. Section Ⅳ validates the proposed algorithm by conducting two numerical experiments. Section Ⅴ summarizes the paper.

II. SPARSE BAYESIAN PARAMETER ESTIMATION WITH FIXED DICTIONARY

Suppose the returned complex radar signal is $\boldsymbol{s}(\boldsymbol{t})$. It is decomposed into a finite sum of the weighted returned signal functions, which represent the scattering centers with different parameters. The signal model can be written as:

$$\boldsymbol{s}(\boldsymbol{t}) = \sum_{k=1}^{K} g_k \boldsymbol{f}(\boldsymbol{r}_k, \boldsymbol{t}) + \boldsymbol{\varepsilon}(\boldsymbol{t}) \tag{1}$$

where $\boldsymbol{t}$ is the time series with length of N; $k = 1,2,\cdots,K$, with K denoting the number of the scattering centers; $g_k, k = 1,2,\cdots,K$ denotes the unknown complex amplitude of the k th scattering center; $\boldsymbol{f}(\boldsymbol{r}_k, \boldsymbol{t}), k = 1,2,\cdots,K$ denotes the parameterized returned signal function of the k th scattering center; $\boldsymbol{r}_k$ are the unknown parameters of the signal model, whose values vary with the difference of the scattering centers; and $\boldsymbol{\varepsilon}(\boldsymbol{t})$ represents the noise.

Then, the signal model in (1) can be expressed in vector-matrix form as:

$$\boldsymbol{s} = \mathbf{D}(\boldsymbol{r}) \cdot \boldsymbol{g} + \boldsymbol{\varepsilon} \tag{2}$$

978-1-4799-6239-6/14 $31.00 © 2014 IEEE
DOI 10.1109/CIT.2014.17

where $\boldsymbol{s}=[s_1,s_2,\cdots s_N]^{\mathrm{T}}$, $\boldsymbol{r}=[\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots \boldsymbol{r}_K]^{\mathrm{T}}$, $\boldsymbol{g}=[g_1,g_2,\cdots,g_K]^{\mathrm{T}}$, $\boldsymbol{\varepsilon}=[\varepsilon_1,\varepsilon_2,\cdots,\varepsilon_N]^{\mathrm{T}}$, and $\mathbf{D}(\boldsymbol{r})=[\boldsymbol{f}(\boldsymbol{r}_1),\boldsymbol{f}(\boldsymbol{r}_2),\cdots,\boldsymbol{f}(\boldsymbol{r}_K)]$.

The problem of interest herein is to estimate the signal parameters $\{\boldsymbol{r}_k\}_{k=1}^K$, amplitudes $\{g_k\}_{k=1}^K$, and K .

A. Basic Model Construction

Since the parameters of $\boldsymbol{s}$ are not known apriori, it is necessary to present a dictionary whose columns are basis functions localized by a series of different initial parameters $\boldsymbol{\theta}=[\boldsymbol{\theta}_1,\boldsymbol{\theta}_2,\cdots,\boldsymbol{\theta}_P]$, where $P \gg K$, $\boldsymbol{r}_k$ and $\boldsymbol{\theta}_p$ represent the different values of the same parameters. Here our goal is to estimate $\{\boldsymbol{r}_k\}_{k=1}^K$ from $\{\boldsymbol{\theta}_p\}_{p=1}^P$. The initial parameterized dictionary is constructed as follows:

$$\mathbf{A}(\boldsymbol{\theta})=[\boldsymbol{f}(\boldsymbol{\theta}_1),\boldsymbol{f}(\boldsymbol{\theta}_2),\cdots,\boldsymbol{f}(\boldsymbol{\theta}_P)] \tag{3}$$

Therefore, the signal $\boldsymbol{s}$ can be sparsely rendered in the dictionary $\mathbf{A}$:

$$\boldsymbol{s}=\mathbf{A}\cdot(\boldsymbol{w}\odot \boldsymbol{z})+\boldsymbol{\varepsilon} \tag{4}$$

where $\odot$ represents the Hadamard product of two vectors, $\boldsymbol{w}=[w_1,w_2,\cdots,w_P]^{\mathrm{T}}$ is the coefficients of $\boldsymbol{s}$ with respect to $\mathbf{A}$, $\boldsymbol{z}=[z_1,z_2,\cdots,z_P]^{\mathrm{T}}$ is used to achieve the sparseness, and only focuses on a small subset of the columns of $\mathbf{A}$. Consequently, the task of parameter estimation can be regarded as recovering the sparse vector $\boldsymbol{z}$ from $\boldsymbol{s}$. After recovering $\boldsymbol{z}$, we search the parameterized basis functions corresponding to the nonzero elements of $\boldsymbol{z}$, therefore, the estimated parameters are obtained.

In Eq. (4), $\boldsymbol{\varepsilon}$ is the independent noise which is assumed to be zero-mean Gaussian distributed with variance $\tau^{-1}\boldsymbol{I}_N$ [12], thus the likelihood function is:

$$\begin{aligned}&\mathrm{p}(\boldsymbol{s}\,|\,\boldsymbol{w},\boldsymbol{z},\tau)=\mathrm{CN}(\boldsymbol{s}\,|\,\mathbf{A}(\boldsymbol{w}\odot \boldsymbol{z}),\tau^{-1}\boldsymbol{I}_N)\\&=\frac{1}{\pi^N\left|\tau^{-1}\boldsymbol{I}_N\right|}\exp(-(\boldsymbol{s}-\mathbf{A}(\boldsymbol{w}\odot \boldsymbol{z}))^{\mathrm{H}}(\tau^{-1}\boldsymbol{I}_N)^{-1}(\boldsymbol{s}-\mathbf{A}(\boldsymbol{w}\odot \boldsymbol{z})))\end{aligned} \tag{5}$$

where $\mathrm{CN}(\cdot)$ specifies the complex Gaussian distribution.

A zero-mean complex Gaussian prior is defined on $\boldsymbol{w}$ [12]:

$$\mathrm{p}(\boldsymbol{w}\,|\,\alpha)=\mathrm{CN}(\boldsymbol{w}\,|\,0,\alpha^{-1}\boldsymbol{I}_P) \tag{6}$$

To complete the specification of the hierarchical prior, Gamma priors [12] are placed on α and the noise precision τ :

$$\begin{aligned}&\mathrm{p}(\alpha)=\mathrm{Gamma}(\alpha\,|\,a,b)=\Gamma(a)^{-1}b^a\alpha^{a-1}\mathrm{e}^{-b\alpha}\\&\mathrm{p}(\tau)=\mathrm{Gamma}(\tau\,|\,c,d)\end{aligned} \tag{7}$$

with $\Gamma(a)=\int_0^\infty t^{a-1}\mathrm{e}^{-t}\mathrm{d}t$ denoting the gamma function. Here we set $a=b=c=d=10^{-6}$.

$\boldsymbol{z}$ is a binary vector with $z_p\in\{0,1\}$. Therefore, setting $z_p=0$ is equivalent to directly removing the p th basis function from $\mathbf{A}$. For $\boldsymbol{z}$, we use the Bernoulli-Beta prior, which is a finite approximation of the Beta process [13]:

$$\begin{aligned}&\mathrm{p}(\boldsymbol{z}\,|\,\boldsymbol{u})=\prod_{p=1}^P \mathrm{Bernoulli}(z_p\,|\,u_p)=\prod_{p=1}^P u_p^{\,z_p}(1-u_p)^{1-z_p}\\&\mathrm{p}(\boldsymbol{u})=\prod_{p=1}^P \mathrm{Beta}(u_p\,|\,\frac{e}{D},\frac{f(D-1)}{D})\end{aligned} \tag{8}$$

where $\mathrm{Bernoulli}(\cdot)$ and $\mathrm{Beta}(\cdot)$ denote Bernoulli and Beta distributions respectively, $\boldsymbol{u}=[u_1,u_2,\cdots u_p]^{\mathrm{T}}$ and $e,f>0$. Usually, we set D to a large, but finite number. Therefore the expectation of u_p , i.e. $\left\langle u_p\right\rangle=\dfrac{e/D}{e/D+f(D-1)/D}$, is close to zero, which means $p(z_p=1)$ is very small and favors sparsity in $\boldsymbol{z}$ (most elements are 0).

B. Variational Bayesian Inference

Bayesian inference seeks to estimate the posterior distributions of the latent variables $\boldsymbol{\Psi}$, given the observed data $\mathbf{H}$ and hyper-parameters Υ . Here we adopt VB [9, 10, 11] inference as a compromise between accuracy and efficiency.

In VB inference, we seek a variational distribution $\mathrm{q}(\boldsymbol{\Psi})=\prod_f \mathrm{q}(\Psi_f)$ to approximate the true posterior distribution of the latent variables $\mathrm{p}(\boldsymbol{\Psi}\,|\,\mathbf{H},\Upsilon)$. The expression is:

$$\ln \mathrm{p}(\mathbf{H}\,|\,\Upsilon)=\mathrm{L}(\mathrm{q}(\boldsymbol{\Psi}))+\mathrm{KL}(\mathrm{q}(\boldsymbol{\Psi})\,|\,\mathrm{p}(\boldsymbol{\Psi}\,|\,\mathbf{H},\Upsilon)) \tag{9}$$

where $\mathrm{KL}(\mathrm{q}(\boldsymbol{\Psi})\,|\,\mathrm{p}(\boldsymbol{\Psi}\,|\,\mathbf{H},\Upsilon))$ is Kullback-Leibler divergence between $\mathrm{q}(\boldsymbol{\Psi})$ and $\mathrm{p}(\boldsymbol{\Psi}\,|\,\mathbf{H},\Upsilon)$. Since $\mathrm{KL}(\cdot)\ge 0$, it follows that $\mathrm{L}(\mathrm{q}(\boldsymbol{\Psi}))$ is a rigorous lower bound of $\ln \mathrm{p}(\mathbf{H}\,|\,\Upsilon)$. Accordingly, the optimal parameterization of $\mathrm{q}(\Psi_f)$ is obtained by minimizing $\mathrm{KL}(\mathrm{q}(\boldsymbol{\Psi})\,|\,\mathrm{p}(\boldsymbol{\Psi}\,|\,\mathbf{H},\Upsilon))$.

In this paper, $\mathbf{H}$ is $\boldsymbol{s}$, $\boldsymbol{\Psi}$ is $\{w_p,\alpha,\tau,z_p,u_p\}_{p=1}^P$, and Υ is $\{a,b,c,d,e,f\}$. The algorithm iteratively updates $\{\mathrm{q}(\Psi_f)\}$ so that $\mathrm{L}(\mathrm{q}(\boldsymbol{\Psi}))$ approaches to $\ln \mathrm{p}(\mathbf{H}\,|\,\Upsilon)$ until convergence. The VB inference of this model is summarized in Algorithm 1, which starts from a set of initial values Υ_0 , and ς is a small positive constant (we set $\varsigma=10^{-6}$).

Algorithm 1: VB-based Inference

$[\boldsymbol{\Psi},\Upsilon,\mathrm{L}]=\mathrm{VB}(\mathbf{H},\Upsilon_0,\varsigma)$

$\Upsilon\leftarrow\Upsilon_0$; $\mathrm{L}(1)=\int \mathrm{q}(\boldsymbol{\Psi})\ln\dfrac{\mathrm{p}(\mathbf{H},\boldsymbol{\Psi})}{\mathrm{q}(\boldsymbol{\Psi})}\mathrm{d}\boldsymbol{\Psi}$

$iter=0$; $change=1$

While $change>=\varsigma$

$iter=iter+1$

$\mathrm{q}(\boldsymbol{w})=\mathrm{N}(\boldsymbol{\mu}_{\mathbf{w}},\boldsymbol{\Sigma}_{\mathbf{w}})$

$\boldsymbol{\Sigma}_w=(\langle\alpha\rangle\boldsymbol{I}_P+\langle\tau\rangle\langle\boldsymbol{z}\boldsymbol{z}^{\mathrm{T}}\rangle\odot(\mathbf{A}^{\mathrm{H}}\mathbf{A}))^{-1}$, $\boldsymbol{\mu}_w=\langle\tau\rangle\boldsymbol{\Sigma}_w diag(\boldsymbol{z})\mathbf{A}^{\mathrm{H}}\boldsymbol{s}$

% where $\langle\bullet\rangle$ means the posterior expectation

$$\mathrm{q}(\alpha)=\mathrm{Gamma}(\tilde{a},\tilde{b}),\quad \tilde{a}=a+P\,,\tilde{b}=b+\sum_{p=1}^{P}\left\langle w_p{}^2\right\rangle$$

$$\mathrm{q}(\tau)=\mathrm{Gamma}(\tilde{c},\tilde{d})$$

$$\tilde{c}=c+N$$

$$\tilde{d}=d+\boldsymbol{s}^{\mathrm{H}}\boldsymbol{s}-2\boldsymbol{s}^{\mathrm{H}}\mathbf{A}diag\left\langle z\right\rangle\left\langle \boldsymbol{w}\right\rangle+sum(diag((\boldsymbol{w}\boldsymbol{w}^{\mathrm{H}})\odot(\boldsymbol{z}\boldsymbol{z}^{\mathrm{T}})\mathbf{A}^{\mathrm{H}}\mathbf{A}))$$

$$\mathrm{q}(z_p)=\mathrm{Bernoulli}(\frac{\mathrm{p}(z_p=1)}{\mathrm{p}(z_p=0)+\mathrm{p}(z_p=1)})$$

$$\mathrm{p}(z_p=1)\propto\exp(\left\langle\ln(u_p)\right\rangle)$$

$$\times\exp(\left\langle\tau\right\rangle(2\left\langle w_p{}^{\mathrm{H}}\right\rangle\boldsymbol{A}_p{}^{\mathrm{H}}(s-\sum_{h\neq p}z_h\left\langle w_h\right\rangle\boldsymbol{A}_h)-\left\langle w_p{}^2\right\rangle\mathbf{A}^{\mathrm{H}}\mathbf{A}))$$

$$\mathrm{p}(z_p=0)\propto\exp(\left\langle\ln(1-u_p)\right\rangle)$$

% where $\boldsymbol{A}_p$ denotes the p th column of $\mathbf{A}$.

$$\mathrm{q}(u_p)=\mathrm{Beta}(\frac{\tilde{e}_p}{D},\frac{\tilde{f}_p(D-1)}{D})\,,\ \tilde{e}_p=e_p+D\left\langle z_p\right\rangle,\tilde{f}_p=f_p+\frac{D}{D-1}(1-\left\langle z_p\right\rangle)$$

$$\left\langle\ln(u_p)\right\rangle=\psi(\frac{\tilde{e}_p}{D})-\psi(\frac{\tilde{e}_p+\tilde{f}_p(D-1)}{D})$$

$$\left\langle\ln(1-u_p)\right\rangle=\psi(\frac{\tilde{f}_p(D-1)}{D})-\psi(\frac{\tilde{e}_p+\tilde{f}_p(D-1)}{D})$$

% where $\psi(\cdot)$ represents the digamma function.

$$\mathrm{L}(iter+1)=\int\mathrm{q}(\boldsymbol{\Psi})\ln\frac{\mathrm{p}(\mathbf{H},\boldsymbol{\Psi})}{\mathrm{q}(\boldsymbol{\Psi})}\mathrm{d}\boldsymbol{\Psi}$$

$$change=(\mathrm{L}(iter+1)\text{-}\mathrm{L}(iter))/\mathrm{L}(iter)$$

End

At convergence of the inference, the posterior means $\langle z\rangle$ and $\langle \boldsymbol{w}\rangle$ can be obtained. We can select the columns of $\mathbf{A}$ via $\{\langle z_p\rangle\}_{p=1}^{P}$. If $\mathrm{q}(z_p=1)\geq 0.5$, then $z_p=1$, and the corresponding column of $\mathbf{A}$ and element of $\boldsymbol{w}$ are selected, otherwise, $z_p=0$, they can thus be pruned. In addition, we can obtain the noise level through the posterior mean of τ , therefore, the denoising can be realized.

III. Parameter Estimation Technique Based on zoom-Dictionary

For the sparse Bayesian parameter estimation method with a fixed dictionary, if the number of the atoms in the dictionary is too small, it is hard to accurately estimate the parameters, while if it is too large, the computational complexity would be increased. Therefore, we propose a sparse Bayesian model with the zoom-dictionary to accurately and rapidly estimate the parameters.

A. Parameter Estimation Technique with Zoom-Dictionary

We first generate the initial basis functions, whose form is $\boldsymbol{f}(\bullet)$. The corresponding parameters of each basis function are $\{\theta_p{}^1,\theta_p{}^2,\cdots,\theta_p{}^C\}_{p=1}^{P}$, where C represents the number of the estimated parameters. According to the practical prior, we can set the maximum and minimize values of the parameters, i.e. the value extents of the parameters. In addition, the initial parameter search intervals $\Delta\boldsymbol{\theta}=\{\Delta\theta_1,\Delta\theta_2,\cdots,\Delta\theta_C\}$ can be set large to decrease the computation cost. With the initial probable values of $\{\theta_p{}^1,\theta_p{}^2,\cdots,\theta_p{}^C\}_{p=1}^{P}$, we can generate the dictionary $\mathbf{A}$ as shown in Eq. (3) with all the ergodic combinations of every parameter of $\{\theta_p{}^1,\theta_p{}^2,\cdots,\theta_p{}^C\}_{p=1}^{P}$.

The steps of this method is described in detail in Algorithm 2. Firstly, we set the maximum number of iterations.

Algorithm 2: Parameter Estimation Technique with Zoom-Dictionary

Step 1 Set $m=1$, and $\Delta\boldsymbol{\theta}^{(m)}=\Delta\boldsymbol{\theta}$.

Step 2 Use Algorithm 1 to prune the current parameter combinations. Then, reserve the residual energy $E_m=(\mathrm{norm}(\boldsymbol{s}\text{ - }\mathbf{A}\cdot(\langle\boldsymbol{w}\rangle\odot\langle z\rangle)))^2$, the sparse selected parameter combinations $\{\theta_l^1,\ \theta_l^2,\cdots,\ \theta_l^C\}_{l=1}^{L}$, the number of the sparse parameter combinations L ($L\leq P$), and the corresponding coefficients $\{w_l\}_{l=1}^{L}$.

Step 3 If the maximum number of the iterations is satisfied, the iterations are halted; otherwise, perform the Step 4, and set $m=m+1$.

Step 4 Set $\Delta\boldsymbol{\theta}^{(m)}=\dfrac{\Delta\boldsymbol{\theta}^{(m-1)}}{2}$, and utilize the zooming method [5] to obtain the refined parameter combinations. Then, generate the new parameterized dictionary $\mathbf{A}$ according to the new parameter combinations. Go to the Step 2.

Now we describe how to refine the parameter combinations. Here we choose the simple equi-spaced refinement. $C=2$ is taken as an example, and the refinement of one parameter combination $\{\theta_l^1,\ \theta_l^2\}$ is shown in below: perform refinement on three points around θ_l^1 and θ_l^2 , i.e., $[\theta_l^1-\Delta\theta_1^{(m)},\theta_l^1,\theta_l^1+\Delta\theta_1^{(m)}]$ and $[\theta_l^2-\Delta\theta_2^{(m)},\theta_l^2,\theta_l^2+\Delta\theta_2^{(m)}]$ with increment halved to $\dfrac{\Delta\theta_1^{(m-1)}}{2}$ and $\dfrac{\Delta\theta_2^{(m-1)}}{2}$, respectively. Then combine the two parameters: $[\theta_l^1-\Delta\theta_1^{(m)},\ \theta_l^2-\Delta\theta_2^{(m)}]$, $[\theta_l^1-\Delta\theta_1^{(m)},\ \theta_l^2]$, $[\theta_l^1-\Delta\theta_1^{(m)},\ \theta_l^2+\Delta\theta_2^{(m)}]$, $[\theta_l^1,\theta_l^2-\Delta\theta_2^{(m)}]$, $[\theta_l^1,\theta_l^2]$, $[\theta_l^1,\theta_l^2+\Delta\theta_2^{(m)}]$, $[\theta_l^1+\Delta\theta_1^{(m)},\theta_l^2-\Delta\theta_2^{(m)}]$, $[\theta_l^1+\Delta\theta_1^{(m)},\theta_l^2]$, $[\theta_l^1+\Delta\theta_1^{(m)},\theta_l^2+\Delta\theta_2^{(m)}]$.

After accomplishing all the iterations, we choose the parameter estimation result of the iteration that corresponds to the minimum residual energy reserved in step 2 as the optimal one.

B. Weighted Clustering

After implementing the Algorithm, there may be many approximate parameter combinations around a true parameter combination. In order to achieve more precise parameter estimation result than that obtained in part A as far as possible, we cluster the approximate parameter combinations surrounding the true parameter combinations.

Here we adopt a simple clustering method, K -means algorithm [14]. Generally, we put the sparse parameter combinations $\{\boldsymbol{\theta}_l\}_{l=1}^{L}$ obtained in part A to the K -means algorithm. Now we suppose K is given, therefore K -means algorithm returns the cluster indices of each point (the parameter combination before clustering) and the cluster centers (the parameter combinations after clustering). However, rather than the obtained cluster centers, we prefer the weighted cluster centers calculated by the reserved coefficients $\{w_l\}_{l=1}^{L}$ and the cluster indices of each point, which is more precise by utilizing the physical characteristics of the signal. The parameter combinations $\{\boldsymbol{\theta}_l\}_{l=1}^{L}$ are divided into K classes, the parameter combinations and the corresponding coefficients belong to the k th class are $\{\boldsymbol{\theta}_l^{(k)}\}$ and $\{w_l^{(k)}\}$, respectively. Therefore, the weighted cluster center of the k th class is $\boldsymbol{\theta}_k = \frac{\sum |w_l^{(k)}| \boldsymbol{\theta}_l^{(k)}}{\sum |w_l^{(k)}|}$. With the weighted cluster centers, we calculate the coefficients through Algorithm 1, instead of summarizing the coefficients of the within-cluster points.

Now, we present how to determine K . We suppose that if the parameter combinations are clustered correctly, the residual energy E_a calculated through the clustered parameter combinations and corresponding coefficients should not be larger than E_b (the residual energy of the optimal iteration before the clustering), otherwise it should be close to E_b , e.g. $E_a - E_b \le E_b / 15$. Therefore, we can take this supposition as the termination condition to select K . In addition, if the selected K is larger than the true value of K , the supposition is also stand up. So we can go through the integer from small to large, i.e. begin with 1, once the value of selected K satisfy the condition, we can terminate the iteration, and the value of K is obtained. .

IV. EXPERIMENTAL RESULTS

A. Complex Sinusoid Model

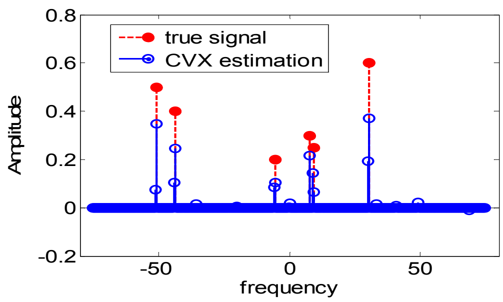
The research of complex sinusoids is of great practical significance. The signal model can be written as:

$$\boldsymbol{s}(\boldsymbol{t}) = \sum_{k=1}^{K} g_k \exp(j2\pi f_k \boldsymbol{t}) + \boldsymbol{\varepsilon}(\boldsymbol{t}) \tag{10}$$

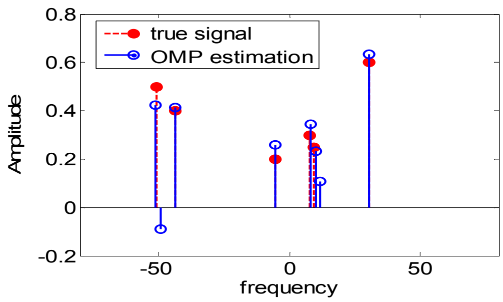
where f_k is the unknown parameter r_k , $\boldsymbol{t} = [0,1,\cdots,N-1]\frac{1}{f_*}$, with N denoting the number of data samples. Here $\boldsymbol{t}$ can be fast time or slow time. Corresponding to the different representations of $\boldsymbol{t}$, N can be the number of the sample points or the pulses, f_* denotes the sampling frequency or the PRF; the parameter f_k indicates the positional information or the Doppler frequency. According to the prior knowledge, we know $\theta_p \in [-f_*/2, f_*/2]$.

Following, a signal with six complex sinusoidal components is used to examine the parameter estimation abilities of two traditional CS methods: CVX[1] , orthogonal matching pursuit (OMP) [15], and our proposed method. We consider $N = 64$ and $f_* = 150\text{HZ}$, therefore, the parameter extent is $\theta_p \in [-75, 75]$. The frequencies of the six complex sinusoidal components are -50.74, -43.73, -5.48, 7.92, 9.34, and 30.35, respectively. The corresponding amplitudes are 0.5, 0.4, 0.2, 0.3, 0.25, and 0.6. For our method, the number of the basis functions is 64, i.e. the initial search interval $\Delta\theta$ is $f_*/64$. However, for another two comparative methods, we adopt the overcomplete dictionary, whose dimensionality is 512, i.e. the initial search interval is $f_*/512$, which is eight times that of our method.

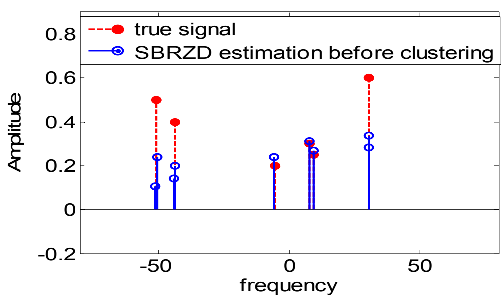
In real applications, the returns are always interfered by strong noise. Therefore, we add the white noise to the signal. SBRZD can estimate the parameters by learning the noise power automatically. However, CVX and OMP cannot solve this problem without knowing some prior knowledge, e.g. the noise power or an appropriate sparsity level of $\boldsymbol{w}$. Therefore, here we assume the noise power is given for CVX and OMP.



(a)

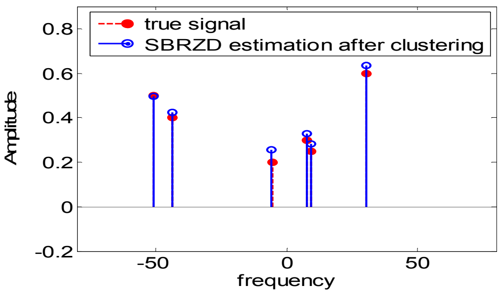


(b)



(c)

[1] The toolbox of CVX is available from http://stanford.edu/~boyd/cvx.



(d)

Fig. 1. Parameter estimation of 10 dB complex sinusoids using CVX, OMP, and SBRZD. (a)results of CVX. (b) results of OMP. (c) results of SBRZD before clustering. (D) results of SBRZD after clustering.

Fig. 1 gives the parameter estimation results of the complex sinusoids under the SNR 10 dB through the four comparative methods. It can be seen that the estimated parameters via SBRZD in Fig. 1(d) are more accurate than that of other comparative methods, which demonstrates the good parameter estimation and denoising capabilities of SBRZD. The parameters estimated through OMP have false points; The result obtained by CVX is similar to our result before clustering in Fig. 1(d), which have several approximate parameters around the true parameters, and this phenomenon conforms to our analysis presented in part B of Section Ⅲ. However, we should pay attention to the fact that the number of the initial basis functions of CVX is further larger than that of SBRZD. Even then, the approximate parameters are more concentrated and accurate in Fig. 1(d) than those in Fig. 1(a), which indicates the utilization of zoom-dictionary is effective. In addition, after clustering, the parameters estimated by our proposed algorithm are more accurate, which confirms the effectiveness and necessity of the weighted clustering.

B. Micro-Doppler signature induced by spinning target

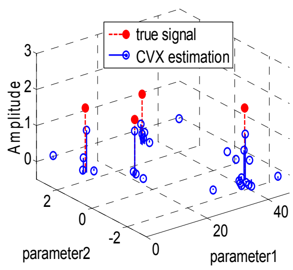
Here we consider another example: the micro-Doppler induced by spinning target [16], which is commonly used in practical applications, e.g. the rotation of a wheel/antenna, Jet Engine Modulation (JEM), the spinning of the space cone-shaped target and so on, whose signal form is:

$$\boldsymbol{s}(\boldsymbol{t}) = \sum_{k=1}^{K} g_k \exp(jb_k \sin(w_s \boldsymbol{t} + e_k)) + \boldsymbol{\varepsilon}(\boldsymbol{t}) \qquad (11)$$

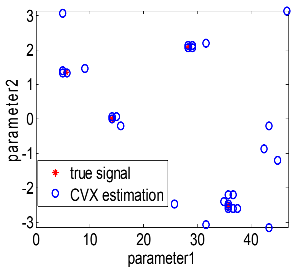
where $\boldsymbol{t} = [0,1,\cdots,N-1]\frac{1}{f_*}$ is the slow time, with N denoting the number of the pulses, f_* denotes the PRF; b_k is a variable which is relative to the radius of the rotation; w_s denotes the frequency of the rotation; e_k represents the initial angle.

Here, we set $w_s = 10\text{rad}/\text{s}$, which can be obtained through other methods in advance, $f_* = 500\text{HZ}$, and $N = 1000$. Therefore, the parameter combinations $\boldsymbol{r}_k = \{b_k, e_k\}$ need to be estimated. We consider $\boldsymbol{s}$ containing $K = 4$ complex spinning components, the parameter combinations $\{\boldsymbol{r}_k\}_{k=1}^{K}$ are $\{35.85, -2.5133\}$, $\{14.24, 0\}$, $\{28.56, 2.0944\}$, and $\{5.61, 1.3464\}$, and the amplitudes $\{g_k\}_{k=1}^{K}$ are 2, 1.6, 1.4, and 1.8. According to the prior knowledge, the parameter extents are $\theta_p^{\,1} \in [0, 47]$ and $\theta_p^{\,2} \in [-\pi, \pi]$. Specially, we still compare our method with CVX, and OMP. For our method, we set $\Delta\theta_1 = 2.5$, $\Delta\theta_2 = 0.2$, and the number of the basis functions is 608. For other methods, we set $\Delta\theta_1 = 2.5/3$, $\Delta\theta_2 = 0.2/3$, the number of the basis functions is 5415, which is almost nine times that of our method.

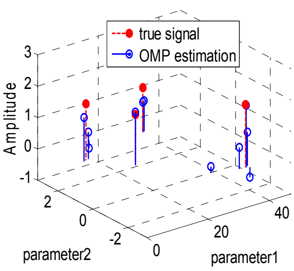
As in experiment A, we also consider the signal with the noise under the SNR 10 dB. The noise power is given for CVX and OMP as well. Fig. 2 gives the parameter estimation results of the four comparative methods. For each method, both 3-D parameter graph (including the amplitudes corresponding to the estimated parameter combinations) and 2-D parameter graph are given. To make the 3-D parameter graphs more clearly, we set a threshold (here is 0.02) to select the larger amplitudes of the basis functions for CVX and BCS. As shown in Fig. 2, CVX and OMP perform badly in parameter estimation with some false parameter combinations. However, inversely, even the SNR is low, our proposed method can not only estimate the parameter combinations accurately, but also obtain the accurate amplitudes corresponding to the parameter combinations.



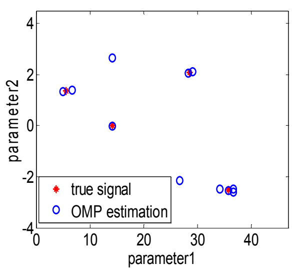
(a)



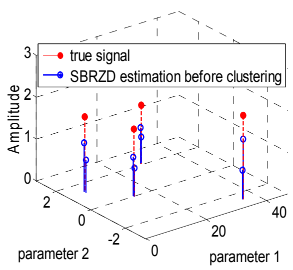
(b)



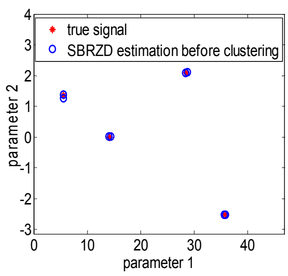
(c)



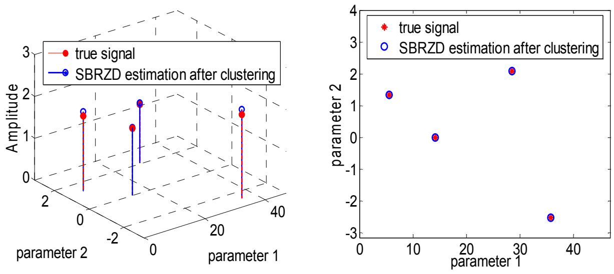
(d)



(e)



(f)



(g) (h)

Fig. 2. Parameter estimation of 10 dB complex Micro-Doppler signature using CVX, OMP, BCS-VB, and SBRZD. (a), (b) results of CVX. (c), (d) results of OMP. (e), (f) results of SBRZD before clustering. (g), (h) results of SBRZD after clustering.

V. CONCLUSION

A novel parameter estimation algorithm which adopts sparse Bayesian representation based on zoom-dictionary is proposed in this paper. The experimental results indicate the parameter estimation capability of SBRZD is superior to other methods especially under the strong noise situation, even though these competitive methods possess better conditions: more initial basis functions and the prior information about noise power.

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